

Extra Practice Qns 2

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Qns 1-2

Anskey:

Section 6.2

1. $f(x) = 4x^2$

$$\begin{aligned}f(s+h) - f(s) &= 4(s+h)^2 - 4s^2 \\&= 4(2s+10h+h^2) - 4(25) \\&= 40h + 4h^2 \text{ (shown)}\end{aligned}$$

$$f'(x) = 8x$$

$$f'(s) = 40$$

2. $f(x+h) = 3(x+h)^2 + 2(x+h) - 1$

$$= 3(x^2 + 2xh + h^2) + 2x + 2h - 1$$

$$= 3x^2 + 6xh + 3h^2 + 2x + 2h - 1$$

$$f(x+h) - f(x) = 3x^2 + 6xh + 3h^2 + 2x + 2h - 1 - 3x^2 - 2x + 1$$

$$= 6xh + 3h^2 + 2h$$

$$\frac{f(x+h) - f(x)}{h} = 6x + 3h + 2$$

$$\text{when } h \rightarrow 0 \quad 3h \rightarrow 0$$

$$6x + 3h + 2 \rightarrow 6x + 2$$

$$f'(x) = 6x + 2$$

$$(b) f'(0) = 2 \quad f'(-2) = -10 \quad f'(8) = 20$$

$$2 = 2(-1) + c \quad c = 4 \quad y = 2x + 4 //$$

3. $D(p) = a - bp$

$$\frac{dD(p)}{dp} = -b$$

4. $C'(x) = p + 2qx$

$$5. \quad f(x+h) = \frac{1}{x+h}$$

$$\begin{aligned} f(x+h) - f(x) &= \frac{1}{x+h} - \frac{1}{x} \\ &= \frac{x - x - h}{x(x+h)} \\ &= -\frac{h}{x(x+h)} \end{aligned}$$

$$\begin{aligned} \frac{f(x+h) - f(x)}{h} &= -\frac{\cancel{h}}{x(x+h)} \times \frac{1}{\cancel{h}} \\ &= -\frac{1}{x(x+h)} \end{aligned}$$

as $h \rightarrow 0$

$$(x+h) \rightarrow x$$

$$-\frac{1}{x(x+h)} \rightarrow -\frac{1}{x^2}$$

$$= -x^{-2} \text{ (shown)}$$

Section 6.6

1 (a) $\frac{dy}{dx} = 0$

(b) $\frac{dy}{dx} = 4x^3$

(c) $\frac{dy}{dx} = 90x^9$

(d) $\frac{dy}{dx} = 0$

2. (a) $2g'(x)$

(b) $-\frac{1}{6}g'(x)$

(c) $\frac{g'(x)}{3}$

3. (a) $6x^5$

(b) $33x^{10}$

(c) $50x^{49}$

(d) $28x^{-8}$

(e) x^{11}

(f) $-2x^{-2}$

$$f'(x) = 4x^{-3}$$

$$(g) \frac{3}{\sqrt[3]{x}} = \frac{3}{x^{1/3}} \\ = 3x^{-1/3}$$

$$f'(x) = -x^{-4/3}$$

$$(h) \frac{-2}{x^{5/2}} = \frac{-2}{x^{5/2}}$$

$$f'(x) = -3x^{-5/2}$$

4 (a) $8\pi r$

(b) $A(b+1)y^b$

$$(c) \frac{1}{A^2 \sqrt{A}} = \frac{1}{A^{2+1/2}} \\ = \frac{1}{A^{5/2}} = A^{-5/2}$$

$$\frac{d}{dA} \left(\frac{1}{A^2 \sqrt{A}} \right) = -\frac{5}{2} A^{-7/2}$$

$$6. F'(x) = x^2$$

$$(a) F(x) = \frac{1}{3}x^3 + c, \text{ where } c \text{ is a constant}$$

$$(b) F'(x) = 2x+3$$

$$F(x) = \frac{2x^2}{2} + \frac{3x}{1} + c, \text{ where } c \text{ is a constant}$$

$$= x^2 + 3x + c$$

$$(c) F'(x) = x^a$$

$$F(x) = \frac{x^{a+1}}{a+1} + c, \text{ where } c \text{ is a constant}$$

$$7 (a) \lim_{h \rightarrow 0} \frac{(s+h)^2 - s^2}{h} = 0$$

$$(b) \lim_{s \rightarrow 0} \frac{(s+1)^3 - 1}{s} = 0$$

$$(c) \lim_{h \rightarrow 0} \frac{5(x+h)^2 + 10 - 5x^2 + 0}{h} = \frac{5(x)^2 - 5x^2}{h} = 0$$

Section 6.7

1 (a) $f'(x) = 1$

(b) $f'(x) = 1 + 2x$

(c) $f'(x) = 15x^4 + 4x^3$

(d) $f'(x) = 24x^3 + x^{-\frac{1}{2}}$

(e) $f'(x) = \frac{1}{2} - 3x + 18x^2$

(f) $f'(x) = -21x^6$

2. (a) $f'(x) = \frac{6}{8}x - 14x^6 - \frac{x}{2}^{-\frac{1}{2}}$

(b) $f(x) = 2x^6 - 3x^2 - x^9 + 1$

$f'(x) = 12x^5 - 4x - 4x^8$

(c) $(x^8 + \frac{1}{x^2})(x^5 + 1) = x^{10} + x^3 + x^4 + \frac{1}{x^2}$

$f'(x) = 10x^9 + 3x^4 + 4x^3 - x^{-2}$

3. (a) $f'(x) = -6x^{-7}$

(b) $f'(x) = (x + x^{-1})x^{\frac{1}{2}}$

$= x^{\frac{3}{2}} + x^{-\frac{1}{2}}$

$= \frac{3}{2}x^{\frac{1}{2}} - \frac{1}{2}x^{-\frac{3}{2}}$

(c) $\frac{1}{\sqrt{x^3}} = \frac{1}{x^{\frac{3}{2}}} = x^{-\frac{3}{2}}$

$f'(x) = -\frac{3}{2}x^{-\frac{5}{2}}$

(d) $f(x) = \frac{x^2 + 1}{x^2 - 1}$

$f'(x) = \frac{(1)(x^2 - 1) - (x^2 + 1)(1)}{(x^2 - 1)^2}$

$= \frac{x^2 - 1 - x^2 - 1}{(x^2 - 1)^2}$

$= -\frac{2}{(x^2 - 1)^2}$

(e) $f'(x) = \frac{(1)(x^5) - (8x^4)(5x+1)}{(x^5)^3}$

$= \frac{x^5 - 8x^4(5x+1)}{x^{15}}$

$= \frac{-4x^5 - 8x^4}{x^{15}} = \frac{-4x - 8}{x^6}$

$$4. (a) \frac{\sqrt{x}-2}{\sqrt{x}+1}$$

$$\begin{aligned} f'(x) &= \frac{(\frac{1}{2}x^{-\frac{1}{2}})(\sqrt{x}+1) - (\sqrt{x}-2)(\frac{1}{2}x^{-\frac{1}{2}})}{(\sqrt{x}+1)^2} \\ &= \frac{\frac{1}{2}x^0 + \frac{1}{2}x^{-\frac{1}{2}} - [\frac{1}{2}x^0 - x^{-\frac{1}{2}}]}{(\sqrt{x}+1)^2} \\ &= \frac{\frac{3}{2}x^{-\frac{1}{2}}}{(\sqrt{x}+1)^2} \\ &= \frac{3}{2\sqrt{x}(\sqrt{x}+1)^2} \end{aligned}$$

$$(b) \frac{x^2-1}{x^2+1}$$

$$\begin{aligned} f'(x) &= \frac{(2x)(x^2+1) - (2x)(x^2-1)}{(x^2+1)^2} \\ &= \frac{2x^3 + 2x - 2x^3 + 2x}{(x^2+1)^2} \\ &= \frac{4x}{(x^2+1)^2} \end{aligned}$$

$$(c) \frac{x^2+2x+1}{x^2-x+1}$$

$$\begin{aligned} f'(x) &= \frac{(2x+1)(x^2-x+1) - (2x-1)(x^2+x+1)}{(x^2-x+1)^2} \\ &= \frac{2x^3 - 2x^2 + 2x + x^2 - x - 1 - (2x^3 + 2x^2 + 2x - x^2 - x - 1)}{(x^2-x+1)^2} \\ &= \frac{\cancel{2x^3} - \cancel{2x^2} - 2x^2 - 2x^2 - \cancel{2x} + \cancel{2x} + x^2 + x^2 - \cancel{x} + \cancel{x} - \cancel{1} + \cancel{1}}{(x^2-x+1)^2} \\ &= \frac{-4x^2 + 2x^2}{(x^2-x+1)^2} \\ &= -\frac{2x^2}{(x^2-x+1)^2} \end{aligned}$$

Section 6.8

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

1. (a) $y = 5u^4$ $u = 1+x^2$

$$\frac{dy}{du} = 20u^3 \quad \frac{du}{dx} = 2x$$

$$\frac{dy}{dx} = 20(1+x^2)^3 \times 2x = 40x(1+x^2)^3$$

(b) $y = u - u^6$ $u = 1 + \frac{1}{x}$

$$\frac{dy}{du} = 1 - 6u^5 \quad \frac{du}{dx} = -x^{-2}$$

$$= 1 - 6\left(1 + \frac{1}{x}\right)^5$$

$$\frac{dy}{dx} = -x^2 + 6x^2 \left(1 + \frac{1}{x}\right)^5$$

2. (a) $Y = -3(v+1)^5$ $v = \frac{1}{3}t^3$

$$\frac{dY}{dv} = -3(5)(v+1)^4 (1) \quad \frac{dv}{dt} = t^2$$

$$= -15(v+1)^4$$

$$= -15\left(\frac{1}{3}t^3 + 1\right)^4$$

$$\frac{dY}{dt} = -15t^2 \left(\frac{1}{3}t^3 + 1\right)^4$$

(b) $K = AL^a$ $L = bt + c$

$$\frac{dK}{dL} = AaL^{a-1} \quad \frac{dL}{dt} = b$$

$$= aA(bt+c)^{a-1}$$

$$\frac{dK}{dt} = abA(bt+c)^{a-1}$$

3(a) $y = (x^2 + x + 1)^{-5}$

$$\frac{dy}{dx} = -5(x^2 + x + 1)^{-6} (2x + 1)$$

$$= -5(2x+1)(x^2+x+1)^{-6}$$

(b) $y = \sqrt{x + \sqrt{x + \sqrt{x}}}$

$$= (x + (x + (x^{\frac{1}{2}}))^{\frac{1}{2}})^{\frac{1}{2}}$$

$$= (x + (x + x^{\frac{1}{2}}))^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{1}{2}(x + (x + x^{\frac{1}{2}}))^{\frac{1}{2}-1} \left(1 + \frac{1}{2}(x + x^{\frac{1}{2}})^{-\frac{1}{2}}(1 + \frac{1}{2}x^{-\frac{1}{2}})\right)$$

(c) $y = x^a (px+q)^b$

$$y = ax^{a-1}(px+q)^b + b(px+q)^{b-1}(p)(x^a)$$

$$= ax^{a-1}(px+q)^b + x^a bp(px+q)^{b-1}$$

$$6. \quad x = b - \sqrt{ap - c}$$

$$= b - (ap - c)^{\frac{1}{2}}$$

$$\frac{dx}{dp} = -\frac{1}{2} (ap - c)^{-\frac{1}{2}} (a)$$

$$= -\frac{1}{2} a (ap - c)^{-\frac{1}{2}}$$

$$7. \quad (a) \quad h(x) = f(x^2)$$

$$h'(x) = f'(x^2) (2x)$$

$$= 2x f'(x^2)$$

$$(b) \quad h(x) = f(x^n g(x))$$

$$= f'(x^n g(x)) [x^n g'(x) + n x^{n-1} g(x)]$$

8. $b(t)$ is the total amount of fuel the car used to travel a certain distance.

$$b'(t) = B'(s(t)), s'(t)$$

$$9. \quad C = 20q - 4q(25 - \frac{1}{2}x)^{\frac{1}{2}}$$

$$\frac{dC}{dx} = -4q(\frac{1}{2})(25 - \frac{1}{2}x)^{-\frac{1}{2}} (-\frac{1}{2})$$

$$= -q(25 - \frac{1}{2}x)^{-\frac{1}{2}}$$

$$10. (a) \quad y = (x^4)^5 \quad y = x^{20}$$

$$\frac{dy}{dx} = 5(x^4)^4 (4x^3) \quad \frac{dy}{dx} = 20x^{19}$$

$$= 5(x^{16}) (4x^3)$$

$$= 20x^{19}$$

$$(b) \quad y = (1-x)^3 \quad y = 1 - 3x + 3x^2 - x^3$$

$$\frac{dy}{dx} = 3(1-x)^2 (-1) \quad \frac{dy}{dx} = -3 + 6x - 3x^2$$

$$= -3(1-x)^2 \quad = -3(x^3 - 2x + 1)$$

$$= -3(1-x)^2$$

Section 6.9

1(a) $f(x) = 5x^4 - 12x^3$

$$f''(x) = 20x^3 - 36x^2$$

1(b) $f(x) = \sqrt{x}$

$$= x^{\frac{1}{2}}$$

$$f'(x) = \frac{1}{2}x^{-\frac{1}{2}}$$

$$f''(x) = -\frac{1}{4}x^{-\frac{3}{2}}$$

1(c) $f(x) = (1+x^2)^{10}$

$$f'(x) = 10(1+x^2)^9(2x) = 20x(1+x^2)^9$$

$$\begin{aligned} f''(x) &= 20(1+x^2)^9 + (20x)9(1+x^2)^8 \\ &= 20(1+x^2)^9 + 180x(1+x^2)^8 \end{aligned}$$

2. $\frac{dy}{dx} = \frac{1}{2}(1+x^2)^{-\frac{1}{2}}(2x)$

$$= x(1+x^2)^{-\frac{1}{2}}$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= (1)(1+x^2)^{-\frac{1}{2}} + (x)(2x)(1+x^2)^{-\frac{3}{2}}(-\frac{1}{2}) \\ &= (1+x^2)^{-\frac{1}{2}} - x^2(1+x^2)^{-\frac{3}{2}} \end{aligned}$$

3(a) $y' = 9x^2 + 2$

$$y'' = 18x$$

(b) $Y' = -4x + 18x^2$

$$Y'' = -4 + 36x$$

$$Y''' = 36$$

(c) $\frac{dz}{dt} = 120 - t^2$

$$\frac{d^2z}{dt^2} = -2t$$

$$\frac{d^3z}{dt^3} = -2$$

$$(d) f(z) = 100z^{-4}$$

$$f'(z) = -400z^{-5}$$

$$f''(z) = 2000z^{-6}$$

$$f'''(z) = -12000z^{-7}$$

$$f^{(4)}(z) = 84000z^{-8}$$

$$\text{when } z = 1$$

$$f^{(4)}(1) = 84000$$

$$4. \quad g(t) = \frac{t^2}{t+1}$$

$$g'(t) = \frac{(t+1)(2t) - (t^2)(1)}{(t+1)^2}$$

$$= \frac{2t^2 - 2t - t^2}{(t+1)^2}$$

$$= \frac{t^2 - 2t}{(t+1)^2}$$

$$g''(t) = \frac{(2t-2)(t+1)^2 - (t^2-2t)(t+1)(2)}{(t+1)^4}$$

$$g''(2) = 2$$

$$5. \quad y' = f'(x)g(x) + g'(x)f(x)$$

$$y'' = f''(x)g(x) + \overset{1}{g'(x)f'(x)} + \overset{2}{f''(x)g'(x)} + \overset{3}{g''(x)f(x)} + \overset{4}{g'(x)f'(x)}$$

$$y''' = f'''(x)g(x) + \overset{1}{g'(x)f''(x)} + \overset{2}{g''(x)f'(x)} + \overset{3}{f'''(x)g(x)} + \overset{4}{f''(x)g'(x)} + \overset{5}{g'''(x)f(x)} + \overset{6}{f'(x)g''(x)} + \overset{7}{g''(x)f'(x)}$$

$$= f'''(x)g(x) + 3f''(x)g'(x) + 3f'(x)g''(x) + g'''(x)f(x)$$

$$6. \quad L = \frac{1}{\sqrt{2t-1}}$$

$$= (2t-1)^{-\frac{1}{2}}$$

$$\frac{dL}{dt} = -\frac{1}{2}(2t-1)^{-\frac{3}{2}}(2)$$

$$= -(2t-1)^{-\frac{3}{2}}$$

$$\frac{d^2L}{dt^2} = \frac{3(2t-1)^{-\frac{5}{2}}}{2}$$

Section 6.10 Exponential Functions

$$f(x) = e^x \Rightarrow f'(x) = e^x$$

$$f(x) = e^{g(x)} \Rightarrow f'(x) = g'(x) e^{g(x)}$$

$$1. (a) \frac{dy}{dx} = e^x + 2$$

$$(b) \frac{dy}{dx} = 5e^x - 9x$$

$$(c) \frac{dy}{dx} = \frac{(1)(e^x) + (x)(e^x)}{(e^x)^2}$$

$$= \frac{e^x(1+x)}{(e^x)^2}$$

$$= \frac{1+x}{e^x}$$

$$(d) \frac{dy}{dx} = \frac{(e^x+1)(1+2x) + (e^x)(x+x^2)}{(e^x+1)^2}$$

$$= \frac{e^x + 2x \cdot e^x + 1 + 2x + x \cdot e^x + x^2 \cdot e^x}{(e^x+1)^2}$$

$$= \frac{2x \cdot e^x + e^x + x^2 \cdot e^x + 1 + 2x}{(e^x+1)^2}$$

$$(e) \frac{dy}{dx} = -1 - e^x$$

$$(f) \frac{dy}{dx} = 3x^2 \cdot e^x + e^x \cdot x^3$$

$$= e^x (3x^2 + x^3)$$

$$(g) \frac{dy}{dx} = e^x \cdot x^{-2} + (-2x^{-3}) \cdot e^x$$

$$= e^x (x^{-2} - 2x^{-3})$$

$$(h) \frac{dy}{dx} = 2(x+e^x) \cdot (1+e^x)$$

$$2(a) \frac{dx}{dt} = (b+2ct)e^t + e^t(a+bt+ct^2)$$

$$(b) \frac{dx}{dt} = \frac{(te^t)(3qt^2) - (p+qt^2)(1 \cdot e^t + t \cdot e^t)}{(te^t)^2}$$

$$= \frac{te^t \cdot 3qt^2 - (p \cdot e^t + pt \cdot e^t + qt^2 \cdot e^t + qt^2 \cdot e^t)}{(te^t)^2}$$

$$= \frac{te^t \cdot 3qt^2 - p \cdot e^t + pt \cdot e^t + qt^2 \cdot e^t + qt^2 \cdot e^t}{(te^t)^2}$$

$$(c) x = \frac{(at+bt^2)^2}{e^t}$$

$$= \frac{e^t (at+bt^2)(2)(a+2bt) - e^t (at+bt^2)^2}{(e^t)^2}$$

$$= \frac{(at+bt^2)(2a+4bt-at-bt^2)}{e^t}$$

$$= \frac{(at+bt^2)(2a+4bt-at-bt^2)}{e^t}$$

$$3(a) \frac{dy}{dx} = (-3) e^{-3x}$$

$$= -3e^{-3x}$$

$$\frac{d^2y}{dx^2} = -3e^{-3x}$$

$$(b) \frac{dy}{dx} = 2(3x^2)(e^{x^3})$$

$$= 6x^2 e^{x^3}$$

$$\frac{d^2y}{dx^2} = (12x)(e^{x^3}) + (6x^2)(3x^2)(e^{x^3})$$

$$= 6x e^{x^3} (2 + 3x^3)$$

$$(c) \frac{dy}{dx} = (-x^{-2}) \cdot e^{\frac{1}{x}}$$

$$\frac{d^2y}{dx^2} = (2x^{-3}) \cdot e^{\frac{1}{x}} + (e^{\frac{1}{x}} \cdot -x^{-3}) \cdot (-x^{-2})$$

$$= e^{\frac{1}{x}} (2x^{-3} + x^{-4})$$

$$(d) \frac{dy}{dx} = 5(4x-3)e^{2x^2-8x+1}$$

$$= 20xe^{2x^2-8x+1} - 15e^{2x^2-8x+1}$$

$$\frac{d^2y}{dx^2} = (20 \cdot e^{2x^2-8x+1} + 20x \cdot 5(4x-3) \cdot e^{2x^2-8x+1}) - 15(6)(4x-3)e^{2x^2-8x+1}$$

$$= e^{2x^2-8x+1} (20 + 20x \cdot 5(4x-3)) - 300x \cdot e^{2x^2-8x+1} + 225e^{2x^2-8x+1}$$

$$= e^{2x^2-8x+1} (20 + 400x - 300 - 300x + 225)$$

$$= e^{2x^2-8x+1} (100x - 55)$$

$$a^x = a^x \ln a$$

$$6.(a) \frac{d}{dx} (e^{e^x})$$

$$= e^x \cdot e^{(e^x)}$$

$$(b) \frac{d}{dt} (e^{t/2} + e^{-t/2})$$

$$= \frac{1}{2} e^{t/2} - \frac{1}{2} e^{-t/2}$$

$$(c) \frac{d}{dt} \left(\frac{1}{e^t + e^{-t}} \right)$$

$$= \frac{d}{dt} (e^t + e^{-t})^{-1}$$

$$= -(e^t + e^{-t})^{-2} \cdot (e^t - e^{-t})$$

$$(d) \frac{d}{dz} (e^{z^3} - 1)^{1/3} = 3z^2 \cdot e^{z^3} \cdot (e^{z^3} - 1)^{-2/3} \left(\frac{1}{3} \right)$$

$$= z^2 \cdot e^{z^3} (e^{z^3} - 1)^{-2/3}$$

7. (a) $y = 5^x$ $a^x = a^x \ln a$

$$\frac{dy}{dx} = 5^x \ln 5$$

(b) $y = x 2^x$

$$\frac{dy}{dx} = (1)(2^x) + (x)(2^x \ln 2)$$

$$= 2^x (1 + x \ln 2)$$

(c) $y = x^2 2^{x^2}$

$$\frac{dy}{dx} = 2x(2^{x^2}) + x^2 (\ln 2 \cdot 2^{x^2} \cdot 2x)$$

$$= 2^{x^2} (2x + x^2 \ln 2 \cdot 2x)$$

$$= 2^{x^2} \cdot 2x (1 + x^2 \ln 2)$$

(d) $y = e^x 10^x$

$$\frac{dy}{dx} = e^x 10^x + e^x (10^x \ln 10)$$

$$= e^x 10^x (1 + \ln 10)$$

Section 6.11 Logarithmic Functions

$$e^{\ln x} = x$$

$$\ln x \text{ derivative} = \frac{1}{x}$$

$$\ln f(x) \Rightarrow \frac{f'(x)}{f(x)}$$

$$1(a) \frac{dy}{dx} = \frac{1}{x} + 3$$

$$(b) \frac{dy}{dx} = 2x - \frac{2}{x}$$

$$(c) \frac{dy}{dx} = 3x^2 \ln x + \frac{1}{x} (x^3)$$

$$= x^2 (3 \ln x + 1)$$

$$(d) \frac{dy}{dx} = \frac{(\frac{1}{x})(x) - (\ln x)(1)}{x^2}$$

$$= \frac{1 - \ln x}{x^2}$$

$$2. (a) y = x^3 (\ln x)^2$$

$$\frac{dy}{dx} = 3x^2 (\ln x)^2 + x^3 \cdot 2 \ln x \left(\frac{1}{x}\right)$$

$$= 3x^2 (\ln x)^2 + 2x^2 \ln x$$

$$(b) y = \frac{x^2}{\ln x}$$

$$\frac{dy}{dx} = \frac{2x \cdot \ln x - \frac{1}{x} \cdot x^2}{(\ln x)^2}$$

$$= \frac{2x \ln x - x}{(\ln x)^2}$$

$$(c) y = (\ln x)^{10}$$

$$\frac{dy}{dx} = 10 (\ln x)^9 \cdot \frac{1}{x}$$

$$= \frac{10 (\ln x)^9}{x}$$

$$(d) y = (\ln x + 3x)^2$$

$$\frac{dy}{dx} = 2 (\ln x + 3x) \left(\frac{1}{x} + 3\right)$$

$$3. (a) \ln(\ln x)$$

$$f'(x) = \frac{\frac{1}{x}}{\ln x}$$

$$= \frac{1}{x \ln x}$$

(b) $\ln \sqrt{1-x^2}$

$$f'(x) = \frac{\frac{1}{2}(1-x)^{-\frac{1}{2}}(-2x)}{\ln \sqrt{1-x^2}}$$

$$= \frac{-x(1-x)^{-\frac{1}{2}}}{\ln \sqrt{1-x^2}}$$

(c) $e^x \ln x$

$$f'(x) = e^x \ln x + e^x \frac{1}{x}$$

(d) $e^{x^3} \ln x^2$

$$f'(x) = \ln x^2 \cdot 3x^2 \cdot e^{x^3} + e^{x^3} \cdot \frac{2x}{x^2}$$

$$= e^{x^3} \left(3x^2 \ln x^2 + \frac{2}{x} \right)$$

(e) $f'(x) = \frac{e^x}{e^x + 1}$

(f) $\ln(x^2 + 3x - 1)$

$$f'(x) = \frac{2x+3}{x^2+3x-1}$$

(g) $2(e^x - 1)^{-1}$

$$f'(x) = \frac{-2(e^x - 1)^{-2} \cdot e^x}{2(e^x - 1)^{-1}}$$

(h) $e^{2x^2 - x}$

$$f'(x) = (4x - 1) \cdot e^{2x^2 - x}$$

Section 7.1 Implicit Differentiation

1. $6x + 2 \frac{dy}{dx} = 0$

$$2 \frac{dy}{dx} = -6x$$

$$\frac{dy}{dx} = -3x$$

Check

$$3x^2 + 2y = 5$$

$$2y = 5 - 3x^2$$

$$y = \frac{5}{2} - \frac{3}{2}x^2$$

$$y' = -3x$$

2. $x^2y = 1$

$$2x \cdot y + x^2 \cdot \frac{dy}{dx} = 0$$

$$x^2 \frac{dy}{dx} = -2xy$$

$$\frac{dy}{dx} = \frac{-2xy}{x^2}$$

$$= -\frac{2y}{x} = -2x^{-3}$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{(-2 \frac{dy}{dx})(x) - (1)(-2y)}{(x)^2} = \frac{-2(-2x^{-3})(x) - (1)(-\frac{2}{x^3})}{x^2} \\ &= \frac{4x^{-2} + 2x^{-3}}{x^2} = 6x^{-4} \end{aligned}$$

Check

$$x^2y = 1$$

$$y = \frac{1}{x^2}$$

$$\frac{dy}{dx} = -2x^{-3}$$

$$= -2x^{-1} \left(\frac{1}{x^2} \right)$$

$$= -2x^{-3}y$$

$$= \frac{-2y}{x}$$

$$\frac{d^2y}{dx^2} = 6x^{-4}$$

$$3(a) \quad x - y + 3xy = 2$$

$$1 - \frac{dy}{dx} + 3 \cdot y + 3x \cdot \frac{dy}{dx} = 0$$

$$1 - \frac{dy}{dx} + 3y + 3x \frac{dy}{dx} = 0$$

$$(3x+1) \frac{dy}{dx} = -1-3y$$

$$\frac{dy}{dx} = \frac{-1-3y}{3x+1}$$

$$3(b) \quad y^5 = x^6$$

$$5y^4 \frac{dy}{dx} = 6x^5$$

$$\frac{dy}{dx} = \frac{6x^5}{5y^4}$$

$$4. \quad u^2 + uv - v^3 = 0 \quad \text{--- (1)}$$

$$2u^2 + v + u \frac{dv}{du} - 3v^2 \frac{dv}{du} = 0$$

$$2u^2 + v + (u - 3v^2) \frac{dv}{du} = 0$$

$$\frac{dv}{du} = \frac{-2u^2 - v}{u - 3v^2}$$

$$\frac{-2u^2 - v}{u - 3v^2} = 0$$

$$-2u^2 - v = 0$$

$$v = -2u^2 \quad \text{--- (2)}$$

$$u^2 + u(-2u^2) - (-2u^2)^3 = 0$$

$$u^2 - 2u^3 - (-8u^6) = 0$$

$$u^2 - 2u^3 + 8u^6 = 0$$

$$u^2 (1 - 2u + 8u^4) = 0$$

$$u = 0 \quad 1 - 2u + 8u^4 = 0 \text{ (no solutions)}$$

when $u=0$

$$v=0$$

$$\therefore (0,0)$$

$$5. \quad 2x^2 + 6xy + y^2 = 18$$

$$4x + 6y + 6x \frac{dy}{dx} + 2y \cdot \frac{dy}{dx} = 0$$

$$4x + 6y + (6x + 2y) \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-4x - 6y}{6x + 2y}$$

when $x=1, y=2$

$$\frac{dy}{dx} = \frac{-4(1) - 6(2)}{6(1) + 2(2)}$$

$$= \frac{-4-12}{6+4}$$

$$= \frac{-16}{10}$$

$$= -1.6$$

$$\frac{dy}{dx} = \frac{-4x-6y}{6x+2y}$$

$$\frac{dy}{dx} = \frac{(-4-6\frac{dy}{dx})(6x+2y) - (-4x-6y)(6+2\frac{dy}{dx})}{(6x+2y)^2}$$

$$= \frac{(-4-6(-1.6))(6+4) - (-4-12)(6-8\cdot 2)}{(6+4)^2}$$

$$= \frac{(\frac{28}{5})(10) - (-16)(-\frac{14}{5})}{100}$$

$$= 1.008$$

6.(a) $x^2+y^2=a^2$

$$2x + 2y \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

(b) $\sqrt{x} + \sqrt{y} = \sqrt{a}$

$$\frac{1}{2}x^{-\frac{1}{2}} + \frac{1}{2}y^{-\frac{1}{2}} \cdot \frac{dy}{dx} = 0$$

$$\frac{1}{2}y^{-\frac{1}{2}} \cdot \frac{dy}{dx} = -\frac{1}{2}x^{-\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{x^{-\frac{1}{2}}}{y^{-\frac{1}{2}}}$$

$$\frac{dy}{dx} = \frac{y^{\frac{1}{2}}}{x^{\frac{1}{2}}}$$

(c) $x^4-y^4=x^2y^2$

$$4x^3 - 4y^3 \frac{dy}{dx} = 2xy^2 + x^2 \cdot 2y \cdot \frac{dy}{dx}$$

$$4x^3 - 2xy^3 - 3x^2y^2 = 4y^3 \frac{dy}{dx} + 2y^2 \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{4x^3 - 2xy^3 - 3x^2y^2}{4y^3 + 2y^2}$$

(d) $e^{xy} - x^2y = 1$

$$(y + x \frac{dy}{dx}) e^{xy} - (2xy + x^2 \frac{dy}{dx}) = 0$$

$$ye^{xy} + e^{xy} \cdot x \frac{dy}{dx} - 2xy - x^2 \frac{dy}{dx} = 0$$

$$(e^{xy} \cdot x - x^2) \frac{dy}{dx} = 2xy - ye^{xy}$$

$$\frac{dy}{dx} = \frac{2xy - ye^{xy}}{e^{xy} \cdot x - x^2}$$

$$7. (a) \quad 2xy \cdot 3y^2 = 9$$

$$2y + 2x \cdot \frac{dy}{dx} + 6y \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2y}{(-2x-2y)}$$

$$\frac{dy}{dx} = \frac{2y}{2x+2y}$$

$$\text{when } x=6, y=1$$

$$\frac{2}{12+2} = \frac{2}{14} = \frac{1}{7}$$

$$y = \frac{1}{7}x + c$$

$$1 = \frac{6}{7} + c$$

$$c = \frac{1}{7}$$

$$\therefore y = \frac{1}{7}x + \frac{1}{7}$$

$$(b) \quad \frac{dy}{dx} = \frac{2y}{2x+2y}$$

$$\frac{d^2y}{dx^2} = \frac{\frac{dy}{dx} \cdot (2x+2y) - (2y)(2+2\frac{dy}{dx})}{(2x+2y)^2}$$

$$= \frac{\frac{1}{7}(12+2) - (2)(2+\frac{2}{7})}{(12+2)^2}$$

$$= -\frac{9}{686} //$$

$$8. (a) \quad xy = g(x^2+y^2)$$

$$y + x \frac{dy}{dx} = g'(x^2+y^2) \cdot 2y \frac{dy}{dx}$$

$$x \frac{dy}{dx} - 2y^2 \frac{dy}{dx} = g'(x^2+y^2) \cdot y$$

$$\frac{dy}{dx} = \frac{g'(x^2+y^2) \cdot y}{x - 2y^2}$$

$$(b) \quad g(x^2+y^2) = x^2 + y^2$$

$$(1 + \frac{dy}{dx}) g'(x^2+y^2) = 2x + 2y \frac{dy}{dx}$$

$$g'(x^2+y^2) \cdot 2x = 2y \frac{dy}{dx} - g'(x^2+y^2) \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{g'(x^2+y^2) \cdot 2x}{2y - g'(x^2+y^2)}$$

$$(c) \quad (xy+1)^2 = g(x^2y)$$

$$2(xy+1) \left(y + x \frac{dy}{dx} \right) = g'(x^2y) (2xy + x^2 \frac{dy}{dx})$$

$$(2xy+2) \left(y + x \frac{dy}{dx} \right) = 2xy \cdot g'(x^2y) + g'(x^2y) \cdot x^2 \frac{dy}{dx}$$

$$2xy^2 + 2y + x(2xy+2) \frac{dy}{dx} = 2xy g'(x^2y) + g'(x^2y) \cdot x^2 \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{(2x^2y + 2x - g'(x^2y)x^2)}{2xy^2 + 2y - 2xy g'(x^2y)}$$

$$9. \quad x^3 F(xy) + e^{xy} = x$$

$$3x^2 \cdot F(xy) + x^3 \cdot F'(xy) \left(y + x \frac{dy}{dx} \right) + \left(y + x \frac{dy}{dx} \right) \cdot e^{xy} = 1$$

$$\text{let } x=1, y=0$$

$$3(1) F(xy) + F'(xy) \left(\frac{dy}{dx} \right) + \left(\frac{dy}{dx} \right) \cdot e = 1$$

$$F(xy) + e = 1$$

$$F(xy) = 1 - e$$

$$3(1-e) + F'(xy) \left(\frac{dy}{dx} \right) + \left(\frac{dy}{dx} \right) e = 1$$

$$(F'(xy) + e) \frac{dy}{dx} = 1 - 3 + 3e$$

$$(F'(xy) + e) \cdot \frac{dy}{dx} = 3e - 2$$

$$\frac{dy}{dx} = \frac{3e - 2}{F'(xy) + e}$$

Section 11.2 Partial Derivatives

1 (a) $\frac{\partial z}{\partial x} = 2$

$$\frac{\partial z}{\partial y} = 3$$

(b) $\frac{\partial z}{\partial x} = 2x$

$$\frac{\partial z}{\partial y} = 3y^2$$

(c) $\frac{\partial z}{\partial x} = 3x^2y^4$

$$\frac{\partial z}{\partial y} = 4x^3y^3$$

2. (a) $\frac{\partial z}{\partial x} = 2x$

$$\frac{\partial z}{\partial y} = 6y$$

(b) $\frac{\partial z}{\partial x} = y$

$$\frac{\partial z}{\partial y} = x$$

(c) $\frac{\partial z}{\partial x} = 20x^3y^2 - 2y^5$

$$\frac{\partial z}{\partial y} = 10x^4y - 10xy^4$$

(d) $\frac{\partial z}{\partial x} = e^{x+y}$

$$\frac{\partial z}{\partial y} = e^{x+y}$$

(e) $\frac{\partial z}{\partial x} = ye^{xy}$

$$\frac{\partial z}{\partial y} = xe^{xy}$$

(f) $\frac{\partial z}{\partial x} = \frac{1}{y}(e^x)$

$$= \frac{e^x}{y}$$

$$\frac{\partial z}{\partial y} = (e^x) \cdot \frac{1}{y^2}$$

$$= -\frac{e^x}{y^2}$$

(g) $\frac{\partial z}{\partial x} = \frac{1}{x+y}$

$$\frac{\partial z}{\partial y} = \frac{1}{x+y}$$

3. (a) $f'_1(x, y) = 7x^6$

$$f'_2(x, y) = -7y^6$$

$$f''_{12} = 0$$

(b) $f'_1(x, y) = (\ln y)(5x^4)$ $f'_2(x, y) = x^5\left(\frac{1}{y}\right)$ $f''_{12}(x, y) = \frac{5x^4}{y}$

$$= 5 \ln y x^4$$

$$= \frac{x^5}{y}$$

$$(c) f'_1(x, y) = 5(x^2 - 2y^2)^4 \cdot 2x$$

$$= 10x(x^2 - 2y^2)^4$$

$$f'_2(x, y) = 5(x^2 - 2y^2)^4 \cdot 4y$$

$$= 20y(x^2 - 2y^2)^4$$

$$f''_{1,2}(x, y) = 10x \cdot 4(x^2 - 2y^2)^3 \cdot -4y$$

$$= -160xy(x^2 - 2y^2)^3$$

$$4.(a) \frac{\partial z}{\partial x} = 3$$

$$\frac{\partial z}{\partial y} = 4$$

$$f''_{1,2}(x, y) = 0$$

$$(b) \frac{\partial z}{\partial x} = 3x^2y^2$$

$$\frac{\partial z}{\partial y} = 2x^2y$$

$$f'_{1,2}(x, y) = 6x^2y$$

$$(c) \frac{\partial z}{\partial x} = 6x^4 - 6xy$$

$$\frac{\partial z}{\partial y} = -3x^2 + 6y^5$$

$$f'_{1,2}(x, y) = -6x$$

$$(d) \frac{\partial z}{\partial x} = (1) \left(\frac{1}{y}\right)$$

$$= \frac{1}{y}$$

$$\frac{\partial z}{\partial y} = x \cdot -\frac{1}{y^2}$$

$$= -\frac{x}{y^2}$$

$$f'_{1,2}(x, y) = -\frac{1}{y^2}$$

$$(e) \frac{\partial z}{\partial x} = 1 \cdot \left(\frac{1}{x+y}\right) + (x-y)(x+y)^2 \cdot -$$

$$= \frac{1}{x+y} - \frac{x-y}{(x+y)^2}$$

$$= \frac{\cancel{x+y} - \cancel{x} + y}{(x+y)^2}$$

$$= \frac{2y}{(x+y)^2}$$

$$\frac{\partial z}{\partial y} = -1 \cdot \left(\frac{1}{x+y}\right) + (x-y)(x+y)^2 \cdot -1$$

$$= \frac{-\cancel{x} - \cancel{y} - x + y}{(x+y)^2} = -\frac{2x}{(x+y)^2}$$

$$f''_{1,2}(x, y) = 2 \cdot \left(\frac{1}{(x+y)^2}\right) + (2y)(x+y)^{-3} \cdot -2$$

$$= \frac{2}{(x+y)^2} - \frac{4y}{(x+y)^3}$$

$$= \frac{2x + 2y - 4y}{(x+y)^3}$$

$$= \frac{2x - 2y}{(x+y)^3}$$

$$5(a) \quad z = x^2 + e^{2y}$$

$$\frac{\partial z}{\partial x} = 2x$$

$$\frac{\partial z}{\partial y} = 2e^{2y}$$

$$\text{second order} = 0$$

$$(b) \quad z = y \ln x$$

$$\frac{\partial z}{\partial x} = \frac{y}{x}$$

$$\frac{\partial z}{\partial y} = \ln x$$

$$\text{second order} = \frac{1}{x}$$

$$(c) \quad z = xy^2 - e^{xy}$$

$$\frac{\partial z}{\partial x} = y^2 - ye^{xy}$$

$$\frac{\partial z}{\partial y} = 2xy - xe^{xy}$$

$$\begin{aligned} \text{second order} &= 2y - [e^{xy} + y(xe^{xy})] \\ &= 2y - e^{xy} + xye^{xy} \end{aligned}$$

$$(d) \quad z = x^y$$

$$\frac{\partial z}{\partial x} = yx^{y-1}$$

$$\ln z = y \ln x$$

$$\frac{1}{z} \frac{\partial z}{\partial y} = \ln x$$

$$\frac{\partial z}{\partial y} = \frac{\ln x}{x^y}$$

$$\begin{aligned} \text{second order} &= \frac{\left(\frac{1}{x}\right)(x^y) - (\ln x)(yx^{y-1})}{(x^y)^2} \\ &= \frac{x^{y-1}(1 - y \ln x)}{(x^y)^2} \end{aligned}$$

$$6.(a) \quad F(S, E) = 2.26 S^{0.44} E^{0.48}$$

$$F'_S(S, E) = 2.26(0.44) S^{-0.56} E^{0.48} = 0.9944 S^{-0.56} E^{0.48}$$

$$F'_E(S, E) = 2.26(0.48) S^{0.44} E^{-0.52} = 1.0848 S^{0.44} E^{-0.52}$$

$$SF'_S(S, E) = 0.9944 S^{0.44} E^{0.48}$$

$$EF'_E(S, E) = 1.0848 S^{0.44} E^{0.48}$$

$$0.9944 S^{0.44} E^{0.48} + 1.0848 S^{0.44} E^{0.48} = 2.0792 F$$

$$\text{where } k = 2.0792$$

$$7. z = (ax+by)^2$$

$$xz'_x = 2(ax+by)(a) \cdot x$$

$$yz'_y = 2(ax+by)(b) \cdot y$$

$$xz'_x + yz'_y = 2(ax+by)a \cdot x + 2(ax+by)b \cdot y$$

$$= 2(ax+by)(ax+by)$$

$$= 2z \quad (\text{Proven})$$

$$8. z = \frac{1}{2} \ln(x^2+y^2)$$

$$\frac{\partial z}{\partial x} = \frac{1}{2} \cdot \frac{dx}{x^2+y^2}$$

$$= \frac{x}{x^2+y^2}$$

$$\frac{\partial^2 z}{\partial x^2} = (1) \left(\frac{1}{x^2+y^2} \right) + (x)(x^2+y^2)^{-2} \cdot (-1) \cdot (2x)$$

$$= \frac{1}{x^2+y^2} - \frac{2x^2}{(x^2+y^2)^2}$$

$$= \frac{x^2+y^2-2x^2}{(x^2+y^2)^2}$$

$$= \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$\frac{\partial z}{\partial y} = \frac{1}{2} \cdot \frac{dy}{x^2+y^2}$$

$$= \frac{y}{x^2+y^2}$$

$$\frac{\partial^2 z}{\partial y^2} = (1) \left(\frac{1}{x^2+y^2} \right) + (y)(2y)(x^2+y^2)^{-2} \cdot (-1)$$

$$= \frac{x^2+y^2-2y^2}{(x^2+y^2)^2}$$

$$= \frac{x^2-y^2}{(x^2+y^2)^2}$$

$$\frac{\partial^2 z}{\partial^2 x} + \frac{\partial^2 z}{\partial^2 y} = \frac{y^2-x^2}{(x^2+y^2)^2} + \frac{x^2-y^2}{(x^2+y^2)^2}$$

$$= \frac{0}{(x^2+y^2)^2}$$

$$= 0 \quad (\text{Shown})$$

$$9. s(x,y) = 2 \ln x + 4 \ln y$$

$$s'_x = \frac{2}{x}$$

$$s'_x(8) = \frac{2}{20} = \frac{1}{10} = 0.1\% \text{ increase in satisfaction}$$

$$s'_y = \frac{4}{y}$$

$$s'_y(30) = \frac{4}{30} = 0.133\% \text{ (3sf)}$$

Section 11.5

$$1. (a) f(-1, 2, 3) = (-1)(2) + (-1)(3) + (2)(3)$$

$$= -2 - 3 + 6 = 1$$

$$f(a+1, b+1, c+1) - f(a, b, c) = (a+1)(b+1) + (a+1)(c+1) + (b+1)(c+1) - (ab + ac + bc)$$

$$= \cancel{ab} + a + b + 1 + \cancel{ac} + a + c + 1 + \cancel{bc} + b + c + 1 - \cancel{ab} - \cancel{ac} - \cancel{bc}$$

$$= 2a + 2b + 2c + 3$$

$$(b) f(tx, ty, tz) = (tx)(ty) + (tx)(tz) + (ty)(tz)$$

$$= t^2 xy + t^2 xz + t^2 yz$$

$$= t^2 (xy + xz + yz)$$

$$= t^2 f(x, y, z) \text{ (shown)}$$

$$2. (a) y = 2.90x_1^{0.015} x_2^{0.250} x_3^{0.350} x_4^{0.408} x_5^{0.030}$$

$$f(2x_1, 2x_2, 2x_3, 2x_4, 2x_5) = 2.90(2x_1)^{0.015} (2x_2)^{0.250} (2x_3)^{0.350} (2x_4)^{0.408} (2x_5)^{0.030}$$

$$= 2.0748y \text{ (Ssf)}$$

$$= 2.07y \text{ (Ssf)}$$